
Automated Component-Level Evaluation of Software Engineering Agents

Kevin Kunkel

mm21lugi@studserv.uni-leipzig.de

Faculty of Computer Science, University of Leipzig

May 5, 2026

ABSTRACT

Current evaluation of software engineering agents predominantly measures end-to-end task success, providing little insight into where and why agents fail. This work investigates whether individual agent steps can be reliably evaluated using LLM-based judges, without requiring human assessment of every decision. We collect trajectories from SWE-agent on SWE-bench instances and parse them into atomic steps, each characterized by the agent's reasoning, tool call, parameters, and observation. We manually annotate a subset of steps across three evaluation dimensions — tool selection, parameter quality, and step coherence — establishing a ground-truth dataset. Using this dataset, we design and systematically compare prompt variants (zero-shot, few-shot, chain-of-thought) for each dimension, measuring agreement between the LLM judge and human labels via accuracy and Cohen's kappa. Our analysis reveals which dimensions of agent behavior are amenable to automatic evaluation, where LLM judges fail systematically, and what this implies for the scalability of glass-box agent evaluation in software engineering contexts.

Keywords Glassbox · Agentic · Development · LLMs

1 Introduction

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequale doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At.

2 Heading: first level

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat.

2.1 Heading: second level

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua quaerat.

2.1.1 Heading: third level

Paragraph Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua quaerat.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua quaerat.

3 Math

Inline: Let a , b , and c be the side lengths of right-angled triangle. Then, we know that: $a^2 + b^2 = c^2$

Block without numbering:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

Block with numbering:

As shown in Equation 1.

$$\sum_{k=1}^n k = \frac{n(n+1)}{2} \tag{1}$$

More information:

- <https://typst.app/docs/reference/math/equation/>

4 Citation

You can use citations by using the `#cite` function with the key for the reference and adding a bibliography. Typst supports BibLateX and Hayagriva.

`#bibliography("bibliography.bib")`

Single citation (Vaswani et al. 2017). Multiple citations (Vaswani et al. 2017; Hinton et al. 2015). In text Vaswani et al. (2017)

More information:

- <https://typst.app/docs/reference/meta/bibliography/>
- <https://typst.app/docs/reference/meta/cite/>

5 Figures and Tables

header 1	header 2
cell 1	cell 2
cell 3	cell 4

Table 1: Lorem ipsum dolor sit amet.



Figure 1: Lorem ipsum dolor sit amet, consectetur adipiscing.

More information

- <https://typst.app/docs/reference/meta/figure/>
- <https://typst.app/docs/reference/layout/table/>

6 Referencing

Figure 1 Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do., Table 1.

More information:

- <https://typst.app/docs/reference/meta/ref/>

7 Lists

Unordered list

- Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do.
- Lorem ipsum dolor sit amet, consectetur adipiscing elit.

Numbered list

1. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do.
2. Lorem ipsum dolor sit amet, consectetur adipiscing elit.
3. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor.

More information:

- <https://typst.app/docs/reference/layout/enum/>
- <https://typst.app/docs/reference/meta/cite/>

Bibliography

- [1] A. Vaswani *et al.*, “Attention is All you Need,” in *NIPS*, 2017.
- [2] G. E. Hinton, O. Vinyals, and J. Dean, “Distilling the Knowledge in a Neural Network,” *ArXiv*, 2015.

APPENDIX A

A.1 Appendix section

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequale doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et.